

Karthik Birre

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SUMMARY

Software Engineer with 4+ years of experience building scalable web applications and .NET microservices using C#, ASP.NET Core, and Angular. Proficient in architecting cloud-native solutions with Microsoft Azure (AKS, Functions, Service Bus, API Management) and optimizing database performance in SQL Server and Cosmos DB. Skilled in secure authentication frameworks including Azure Entra ID, JWT, and Key Vault. Adept at streamlining CI/CD pipelines with Azure DevOps, Docker, and Helm, delivering maintainable, test-driven applications that enhance performance, reliability, and business value.

TECHNICAL SKILLS

Backend: .NET microservices (4-layer: Data, Repository, Service, Controller), ASP.NET Core Web API & MVC, REST APIs, Dapper, Entity Framework Core (EF Core), ADO.NET, AutoMapper, xUnit & Moq testing, Azure Cache, ASP.NET Core HealthChecks, C#, LINQ

Databases: Azure SQL Database (relational), Azure Cosmos DB (NoSQL), MS SQL Server, EF Core migrations, Transact-SQL

Authentication & Security: JWT authentication, Azure Entra ID (AAD), Azure Key Vault, Azure API Management, RBAC

Messaging & Integration: Azure Service Bus, Azure Functions, Azure API Management

Frontend: Angular, TypeScript, JavaScript (ES6+), HTML5, CSS3, Bootstrap, Responsive Design, Web Accessibility (WCAG)

Cloud & DevOps: Microsoft Azure (AKS, App Service, Functions, Blob Storage, Service Bus, Cosmos DB, API Management, Key Vault, Azure Monitor), Azure DevOps YAML pipelines (branch policies, pull request gates), build/release pipelines, Docker & Helm on AKS, CI/CD

Testing & Monitoring: xUnit, NUnit, MSTest, Moq, Selenium, Postman, Serilog, Application Insights, Load Testing, Static Code Analysis

Source Control & Agile: Azure DevOps, Agile/Scrum, User Stories, Sprint Planning, Code Reviews, Documentation, Design Patterns

PROFESSIONAL EXPERIENCE

Software Engineer, Dell Technologies

Aug 2024 – Present | Remote, USA

- Developed ASP.NET Core Web API microservices for global device lifecycle management, handling 15,000 monthly requests and improving service reliability across multiple enterprise product platforms.
- Built Angular and TypeScript portals tightly integrated with .NET APIs, reducing manual warranty claim resolution by 35% and improving efficiency for customer support operations.
- Designed asynchronous messaging pipelines with Azure Service Bus and Functions, automating device replacement workflows and reducing operational delays for over 5,000 monthly transactions.
- Strengthened application security using JWT authentication, Azure Entra ID, and Key Vault, achieving 100% compliance in quarterly enterprise audits and internal Dell security reviews.
- Enhanced SQL Server performance with EF Core optimizations and Dapper queries, reducing data access latency by 28% and improving reporting for 400+ operations managers.
- Automated CI/CD pipelines with Azure DevOps YAML and Docker, cutting deployment cycles by 40% while maintaining 99.7% uptime for mission-critical global applications.
- Introduced repository-service-controller pattern in .NET microservices, improving maintainability, reducing code duplication, and accelerating feature delivery timelines by 25% across multiple development squads.

Software Engineer, HCL Tech

Jun 2020 – Jul 2023 | Hyderabad, India

- Engineered backend modules for a large-scale healthcare claims processing system using ASP.NET Core Web API and EF Core, streamlining adjudication workflows and improving accuracy.
- Designed Angular and TypeScript dashboards for healthcare providers, delivering intuitive claim status visibility that reduced manual inquiries by 30% and accelerated patient reimbursement cycles.
- Implemented Azure Functions and Service Bus integrations to automate real-time eligibility checks, reducing claim validation delays by 40% and ensuring compliance with HIPAA requirements.
- Secured applications with JWT authentication and Azure Key Vault, enforcing strict access policies and preventing unauthorized access across thousands of distributed user accounts.
- Enhanced maintainability by introducing repository-service-controller architecture in .NET microservices, reducing code duplication and enabling faster onboarding of five new developers within three months.
- Established automated testing with xUnit, NUnit, and Moq, increasing coverage to 85% and preventing regression issues during critical healthcare software updates and production releases.
- Improved application performance by implementing Azure Cache and optimizing LINQ queries in .NET microservices, reducing response times by 32% for high-traffic healthcare workflows.

EDUCATION

Master of Science, Marist University

Aug 2023 -May 2025 | NY, USA

Management Information Systems

Bachelor of Technology, Bankatlal Badruka College for Information Technology

Jul 2019 – Jul 2022 | Hyderabad, India

Computer Science